Lesson 13: A Hit That Reacts

Scratch

VIDEO GAME CREATION

This week: getting hit finally LOOKS like getting hit. A brief flash, then a small explosion, before the enemy disappears.

Open your own [YourName]Game project from Lesson 12 - don't start a new one.

Important: your score, lives, wave, and win/lose logic should all still work exactly as before. Today only changes how the enemy's removal LOOKS, not when your game-state actually changes.

Setup

1. Open your [YourName]Game project from Lesson 12.
2. Add the explosion_placeholder.png costume to a brand-new sprite named Explosion (Scratch will let you draw/import a new sprite from a file).
3. Add "set [color] effect to 0" to the Enemy's existing "when I receive [spawn wave]" reset script (or wherever each enemy resets), so a fresh enemy never starts already flashed.

Level 1 - Minimal (MVP)

On every hit, the enemy briefly flashes, and a small independent explosion effect appears and cleans itself up. Score, lives, wave, and win/lose logic are all unaffected.

4. On the Explosion sprite, hide it by default, and add a new script: "when I receive [explosion]" -> show -> go to x/y of Enemy (using the enemy's position at the moment of the broadcast) -> wait 0.2 seconds -> hide.
5. In the Enemy's touching-Bullet branch, right after "change enemies_left by -1" and the existing sound/score lines, add: broadcast [explosion] (this spawns the visual independently, with zero effect on anything else), then "set [color] effect to 100", then "wait 0.1 seconds".
6. Move the existing win-check if/else and "delete this clone" to run AFTER the wait - everything that was already there stays exactly the same, just one step later.
7. Click the stage and play through a full round - each hit should flash the enemy briefly, and a small burst should appear at that spot for a moment.

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Enemy's touching-Bullet branch (revised):
if <touching Bullet?> then
  change score by 10
  change enemies_left by -1
  start sound [sfx_hit]
```

```
broadcast [explosion]
set [color] effect to 100
wait 0.1 seconds
if <enemies_left = 0> then
  if <wave = total_waves> then
    set game_active to 0
    start sound [sfx_win]
    broadcast [you win]
  else
    change wave by 1
    set enemies_left to 5
    start sound [sfx_wave]
    broadcast [spawn wave]
delete this clone
```

Level 2 - Complete (Alpha/Beta)

The flash and explosion play correctly on every single hit, with no leftover explosions, no enemies stuck mid-flash, and no change to score/lives/wave timing.

- Play a full round: every single hit shows a flash and an explosion, with no enemy getting visually "stuck" flashed.
- Confirm score, wave-clear, and win/lose all still happen at exactly the moments they did last week - only the enemy's disappearance is 0.1 seconds later than before.
- Confirm the broadcast-based Explosion sprite never blocks or delays anything else - broadcasts don't wait for their receivers to finish.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Audio/Visual: instead of one flat 0.1-second flash, alternate the color effect a couple of times (e.g. 100, then 0, then 100 again) with short waits between, for a stronger flicker.
- Visual: give the Explosion sprite 2-3 different costumes and cycle through them with "next costume" during its 0.2-second lifetime for a simple hand-animated burst instead of one static image.
- Structure: notice the Explosion sprite knows nothing about score, lives, or waves - it does exactly one job, triggered independently of the rest of the hit-handling logic.

If Something's Wrong

- Enemy vanishes instantly, no flash visible: confirm "wait 0.1 seconds" was actually added, and that the win-check/delete block was moved to AFTER it, not before.
- Explosion never appears: confirm the Explosion sprite actually has a "when I receive [explosion]" script, and that its costume isn't accidentally still hidden.
- Score changes twice per hit, or win-check fires early: confirm the win-check/scoring lines weren't accidentally duplicated when moving them below the wait.

Want More?

Student Resource List > Scratch section, or scratch.mit.edu/ideas